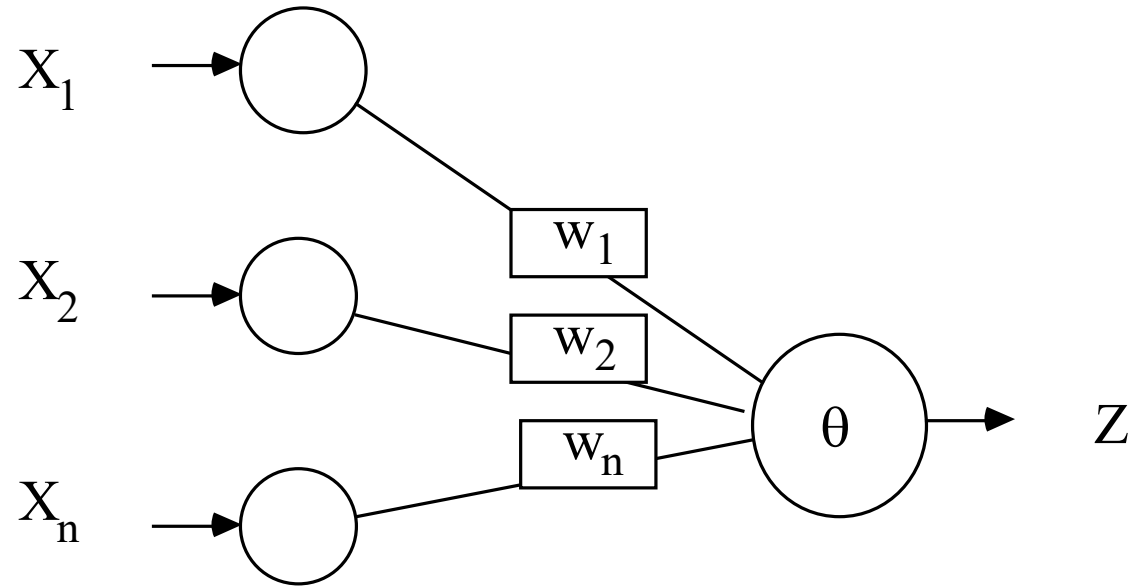


# Do we need to adjust $\theta$ ?

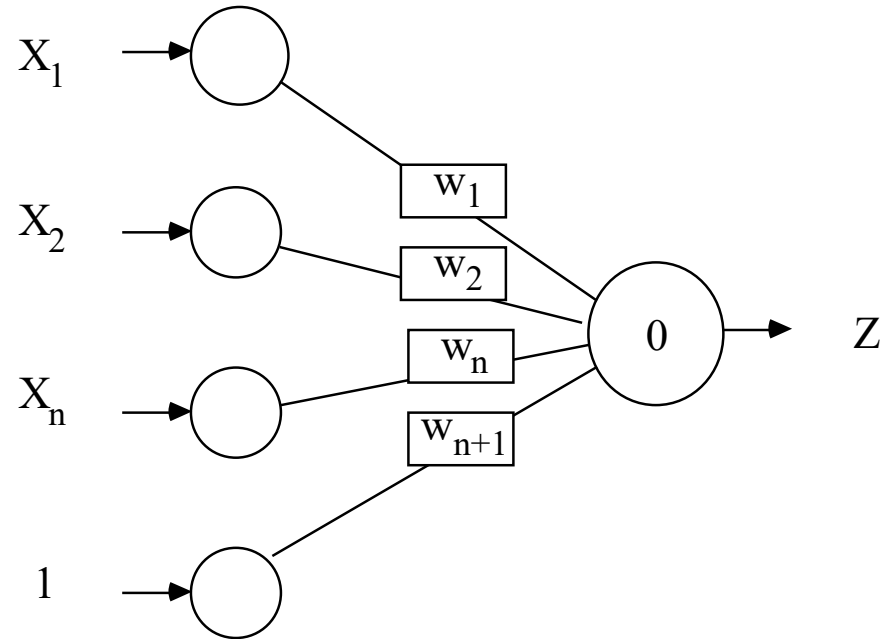
---



Yes, in most cases!

# Treat $\theta$ as an extra weight?

---



where  $w_{n+1} = -\theta$

# Augmented Pattern Vectors

---

1 0 1 -> 0

1 0 0 -> 1

Augmented Version

1 0 1 1 -> 0

1 0 0 1 -> 1

- ❑ Treat threshold like any other weight. No special case. Call it a *bias* since it biases the output up or down.
- ❑ Since we start with random weights anyways, can ignore the  $-\theta$  notion, and just think of the bias as an extra available weight. (note the author uses a -1 input)
- ❑ Always use a bias weight